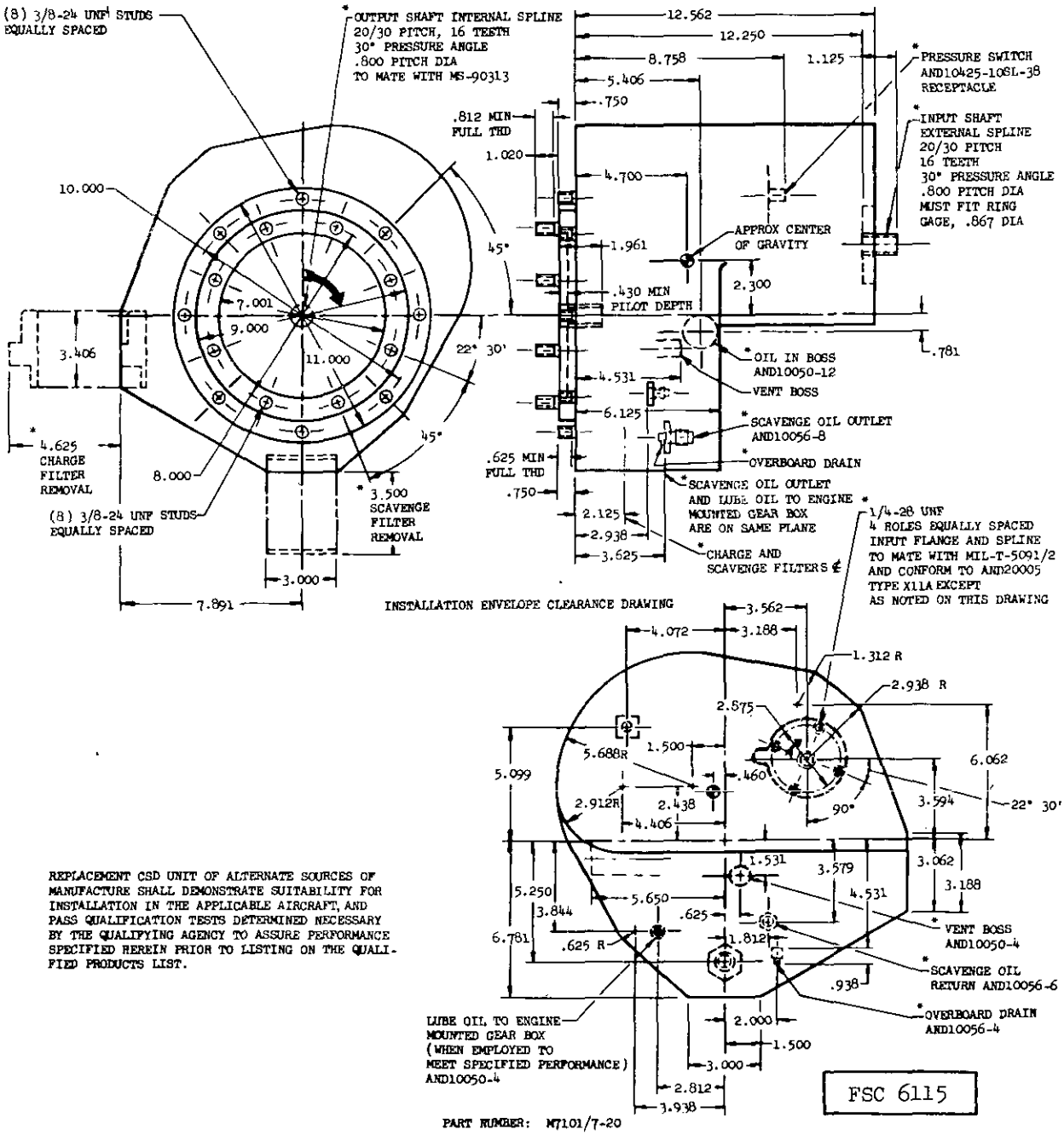
MILITARY SPECIFICATION SHEET 3 August 1965

TRANSMISSION, CONSTANT SPEED, HYDRAULIC 20KVA, S-2/E1 AIRCRAFT

The complete requirements for procuring the transmission described herein shall consist of this document and the issue in effect of MIL-T-7101(WP).

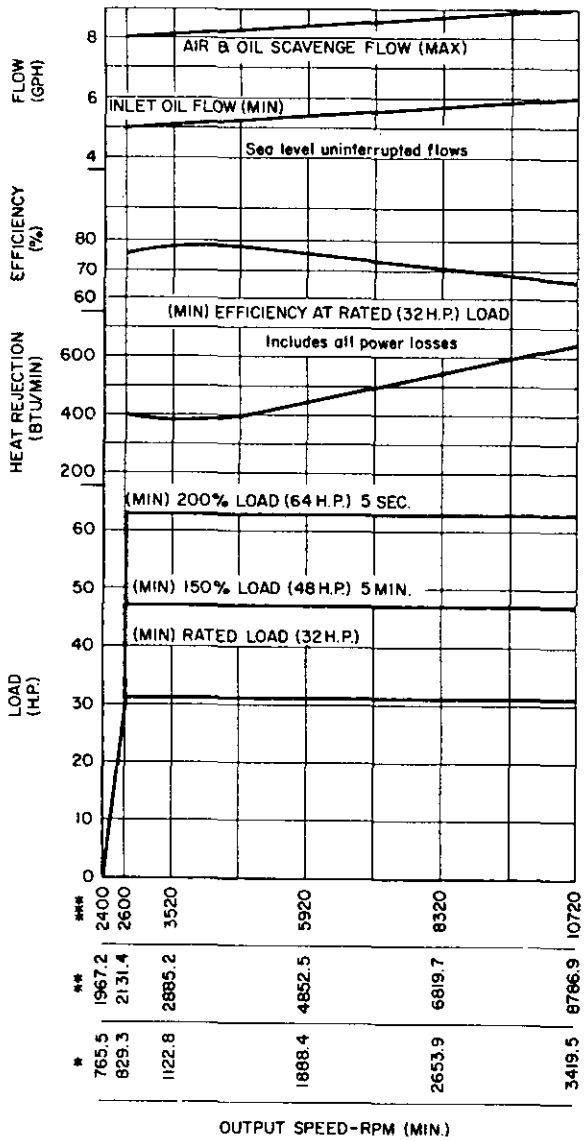


REPLACEMENT CSD UNIT OF ALTERNATE SOURCES OF MANUFACTURE SHALL DEMONSTRATE SUITABILITY FOR INSTALLATION IN THE APPLICABLE AIRCRAFT, AND PASS QUALIFICATION TESTS DETERMINED NECESSARY BY THE QUALIFYING AGENCY TO ASSURE PERFORMANCE SPECIFIED HEREIN PRIOR TO LISTING ON THE QUALIFIED PRODUCTS LIST.

PART NUMBER: M7101/7-20

FSC 6115

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LOAD & SPEED SCHEDULE (500 HOUR) 121.1°C MIN OIL IN

PERCENT OF TOTAL TIME MIN.	INPUT SPEED TO EMGB (RPM)	LOAD (H.P.) MIN.
40%	8500-10720	0-32
35%	5000-8500	16-24
25%	2600-5000	16-32

LOAD & SPEED SCHEDULE (1000 HOUR) 93.3°C OIL IN

INPUT SPEED (RPM)	LOAD (H.P.) MIN.	PERCENT OF TOTAL TIME MIN.
2600-10720	32	30%
2600-10720	16	67%
2600-10720	40	3%

FREQ. DEVIATIONS (SHOCK LOADS)

INPUT SPEED 2700-10720	
LOAD CHANGE PERCENT MIN.	TRANSIENT CPS MAX.
0-100%	±20
0-150%	±32
0-200%	±45

FREQ DEVIATIONS (SPEED CHANGE)

INPUT SPEED 2700-10720	
CHANGE RPM/SEC MIN.	MAX. TRANSIENT CPS MAX.
0-50	±4 CPS
300-600	±9 CPS
900-1200	±10 CPS

NOTES:

1. The transmission output speed shall be 8000 ± 80 RPM for steady state conditions of load and input speed.

2. The *oil inlet temperature to the transmission* shall be within the range of 37.8°C. to 121.1°C. Intermittent oil-in temperature of 135°C. may exist for 5 minute periods not to exceed 5 percent of the total life of a trans. The transmission shall be capable of starting with an oil inlet temperature of -55°C.

3. Minimum inlet oil pressure 7 inches mercury absolute.

4. Minimum internal case pressure 7 inches mercury absolute.

5. Recommended degree of oil filtration 46 micron maximum.

6. The maximum differential case pressure with respect to ambient shall be 16 inches of mercury.

7. The weight of the transmission shall not exceed 65 lbs. dry, exclusive of EMG box (if employed).

8. Transmission overhung moment 306 in. lbs.

9. The overhung moment of the transmission generator package shall not exceed 900 in. lbs.

10. The loading device for the transmission shall have a Wr^2 within the range of 0.200 to 0.700 lbs. $ft.^2$

11. The transmission shall not increase the overall acoustical level more than 5 decibels above the 90 decibel background noise.

12. The transmission shall operate as specified in all attitudes specified when installed in the applicable aircraft under service conditions.

13. Transmission shall be finished in a color conforming to FED-STD-595 Color Number 17875. The finish shall be unaffected by environmental conditions specified herein.

14. Flight Test: 100 hours flight time on the S-2 aircraft is required for qualification.

15. * Engine (crankshaft) RPM
** Engine Generator Pad Output RPM - (w/o EMGB)
*** Engine Mounted Gear Box Output RPM (with EMGB)

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3.2 Operating conditions - Change to read as follows: "The transmission shall perform as specified under each and any natural combination of the following conditions."

3.2.1 Pressure altitude - Change to read: "The transmission shall be insensitive to altitude conditions up to 35,000 feet."

3.2.2 Temperature - Change to read as follows: "The transmission shall operate as specified when the ambient temperature is in the range of -53°C to +121°C."

3.2.7 Operating position - Delete and substitute "The transmission shall be capable of satisfactory operation when mounted with the rotational axis in a horizontal position and provided the inlet oil pressure is maintained above 4 psia (when an external oil reservoir is provided), through aircraft maneuvers as follows:

a. Level position with transmission inclined 30° to either side.

b. 0° to 30° diving angle with up to 10° inclination on either side.

c. 31° to 90° diving angle with up to 10° inclination on either side for a period of 60 seconds.

d. 0° to 45° climbing angle with up to 10° inclination on either side

e. 46° to 90° climbing angle with up to 10° inclination on either side for a period of 60 seconds.

f. Navigating and/or inverted flight for 15 seconds.

3.2.8 Load speed schedule - The transmission shall perform as specified and be capable of providing 100 cycles under the load-speed schedule below with an oil inlet temperature of 93.3°C.

Condition No.	Time (Minutes)	Input Speed (RPM)	Load (KW)
1	58	6900	12
2	2	6900	25
3	58	5860	12
4	2	2100	25
5	58	6600	12
6	2	6900	25
7	58	5860	12
8	2	2100	25
9	58	6900	12
10	2	6900	25
11	29	5860	12

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Condition No.	Time (Minutes)	Input Speed (RPM)	Load (KW)
12	29	2100	25
13	2	2100	15
14	58	3500	12
15	2	6900	25
16	58	3500	12
17	2	2100	25
18	58	6600	12
19	2	6900	25
20	58	5860	12
21	2	2100	25

3.3.3.1 Fluids - Change to read as follows: "The transmission shall use fluid conforming to MIL-L-7808 or MIL-L-23699 as a lubricant and cooling medium."

3.3.3.2 Lubricants - Interface spline connection per MIL-G-21164.

3.3.3.3 Replenishment of fluid lubricant - **Applicable. Relubrication of splines each 120 hours is permitted.**

3.3.6 Torsional vibration - Change to read as follows: "The transmission shall introduce no harmful torsional vibrations between its imput and accessory nor shall it increase the amplitude of any torisonal vibrations applied to the input end."

3.3.7 Torque limiting - The input shear section shall provide for decoupling when torque exceeds 4030 ±240 pound inches.

3.3.8 Leakage - Change to read as follows: "The total overboard fluid leakage of the transmission shall not exceed 2cc per hour per rotating seal while operative and 1 cc per rotating seal **inoperative.** The total leakage from all points other than rotating seals shall not exceed 1 cc per hour operative or inoperative."

3.3.10 Direction of rotation - Change to read as follows: "The input and output rotation shall be as shown on the transmission drawing."

3.3.11 Intermittent overload - Change second sentence to read: "The transmission shall operate as specified herein while carrying 150 percent load minimum once every two (2) hours of operation and 200 percent load minimum once every two (2) hours of operation during a 500 hour life schedule. During a 1000 hour life schedule, the transmission shall operate as specified herein while carrying 150 percent load minimum once every ten (10) hours of operation and 200 percent load minimum once every twenty-five (25) hours of operation."

3.3.11.1 Overheat indicator - Delete. The overheat indicator shall not be provided as a part of this transmission.

3.3.12 Remote speed adjustment - Factory adjustment shall be lock-wired.

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3.3.15 Service life - Change to read: "The transmission shall operate under any natural combination of conditions described in this specification, the useful life without replacement or reprocessing of major parts shall be 1000 hours or more in the aircraft when operated in accordance with the applicable load and speed schedule.

3.7.1 Aluminum alloy parts - In second sentence, change "and from" to read "such as".

3.10.1 Nameplate - Change nameplate information to read as follows:

Transmission, A-C Generator, Constant Speed
Part Number MS Stock Number
Specification MIL-T-7101/7
Manufacturer's Part Number
Manufacturer's Serial Number
Manufacturer's Name
Order Number
Acceptance Stamp
U.S.

3.10.1.1 Manufacturer's part number on nameplate - In first sentence, change "contractor's assembly drawing" to read "contractor's outline drawing".

4.1.3. Color - The exterior surfaces of the transmission (and all auxilliary equipment required to obtain performance specified herein) shall be finished in a color conforming to FED-STD-595 color number 17875.

4.2.2. Qualification tests - Delete "Parallel Operation".

4.3.2 Tests - Delete "Parallel Operation".

4.4.6 Auxilliary equipment - Change to read: "Any additional equipment such as intercoolers, coolant, etc. indicated by the contractor as necessary to meet the requirements of this specification shall be used".

4.6.2.1 High temperature - Change first sentence to read: "The transmission shall be subjected to an ambient temperature of 121 ±2°C for at least 12 hours before operating, except use MIL-E-5272, Procedure 1."

4.6.3 Flight test - The transmission shall be flight tested in the applicable aircraft for 100 hours under conditions specified herein. Evidence of unsatisfactory performance shall be a cause for rejection.

4.6.4.1 Underspeed operation - Change to read: "The transmission shall satisfactorily carry reduced loads below the minimum input speed as specified in the detail specification."

4.6.5 Leakage - Change to read: "The total overboard fluid leakage of the transmission shall not exceed 2 cc per hour per rotating seal while operative and 1 cc per rotating seal inoperative. The total leakage

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from all points other than rotating seals shall not exceed 1 cc per hour operative or inoperative."

4.6.7 Torque limiting - Delete

4.6.8 Overspeed - Change to read: "This test shall be made while the transmission is hot as a result of testing and shall be made at no load. Input speed to the transmission shall be 11,800 RPM (110 percent of maximum rated speed) for a period of 30 seconds. The transmission shall demonstrate its ability to operate at input overspeed conditions without mechanical failure and without impairing subsequent performance. During the above test the transmission output speed shall not exceed 10,000 RPM. In the event of an output overspeed, the transmission output speed shall not exceed 10,000 RPM (500CPS)."

4.6.11 Speed regulation - Part (c) - Change to read as follows: "Speed regulation at 121° ±2°C. - Repeat part 1 through 7 under (a) at 121° ±2°C."

4.6.12.1 Determination of total deviations -

(a) Ambient temperature - Change "71° ±2°C. "to read" 121° ±5°C.

(b) Altitude - Change "65,000 feet" to read "100,000 feet".

4.6.13.1 Speed stability - Change last sentence to read: "Evidence of sustained oscillations other than steady state frequency modulation shall be cause for rejection."

4.6.13.2 Load stability - Change "±4 percent" to read "±6.25 percent". Change last sentence to read: "Evidence of sustained oscillations other than steady state frequency modulation shall be cause for rejection."

4.6.13.3 Add new paragraph: "4.6.13.3 Steady state frequency modulation amplitude - The steady state frequency modulation amplitude shall not exceed 0.25 percent using the following formula:

$$\frac{RPM_{max} - RPM_{min}}{RPM_{max} + RPM_{min}} \times 100 = 0.25 \text{ percent}$$

except in the input speed range of 7500 - 7845 RPM during which the amplitude shall not exceed 0.5 percent."

4.6.16.1 Linear - Change last sentence to read: "The operation of the transmission shall not be affected when these conditions are applied except when accelerations are applied parallel to the rotational axis of the transmission, the output speed deviations shall not exceed the limits of Paragraph 4.6.13.1."

4.6.18 Vibration - Change to read: "The transmission shall be tested in accordance with Procedure I, MIL-E-5272A, except that the entire test be conducted at room ambient."

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4.6.20 Altitude performance - Change to read: "The transmission shall be insensitive to altitude conditions."

5 Preparation for delivery - Delete. Preparation for delivery shall be as specified in the contract.

6.1 Weights and dimensions - Add "Shall be provided to the Qualifying Agency" .

Paragraphs 3.3.12; 3.3.14; 3.3.18; not applicable.

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